Index-calculus relation yield across 48 isogeny classes of $\mathbb{F}_{2^{31}-1}$: no class or curve effect

Phase 3 of `research/fp31_whole_field_census_20261111`, 2026-11-12.

- **Protocol:** `phase3/PROTOCOL.md`, frozen before data at commit `58e501e346`, sha256 `572f7dfb…eeea`.
- **Instrument:** the P2 instrument of `research/isogeny_class_ic_yield_20261102`, unchanged: `p2tool spec` → `ecbench run` → `ecbench verify --replay 4` → `p2tool analyze`. The cross-class tests are computed by `fp31-class-census ic-analyze`.
- **Execution:** task cards compiled by the experiment director and run by the mechanical executor. Attempt 1 was lost to a container restart after 37 of 96 cells. The loss is in `benchmarks/losses/` and described in `phase3/RESTART_NOTE.md`. Attempt 2 resumed with identical bound inputs, and every cell's workload and seeds are fixed.
- **Host:** 4-vCPU cloud container. The route is local, recorded in PROTOCOL §7, and ecbench graded the rows L1/L2.

Classification: a null-hypothesis characterization. It leaves the index-calculus baseline unchanged and adds no leaderboard row. No claim here has a pcc claim id: all are **[pcc:unverified]**.

What ran

- **Population.** The 48 prime-order classes of phase 2, 16 in each endomorphism-ring stratum (S1: one ring, S2: two, S3: three or more). Each class contributes every curve its phase-2 isogeny walk reached: 332 curves, each with the walk's instance point P as generator. `ic-classes` checked P on the curve, $nP = O$ and the j-invariant. Each class also has 2 seeded u-scaled isomorphic models.
- **Measurement.** For every class and $m \in \{3, 2\}$, the arms are `ic.pipeline`, `prime-abscissa` factor base, `mitm:m` oracle. The factor base has the same size on every curve: $|F| = 378$ for $m = 3$ and $|F| = 1448$ for $m = 2$, the table cap, identical in every class. Each curve gets 2 rounds × 2 targets of a fixed number of decomposition attempts, plus a planted arm with factor base $|F| \cdot 1.2^{1/m}$, whose expected yield is +20 %.
- **Totals.** 96 cells, 0 failed. For $m = 3$: 260,773 relations in 7,921,136 attempts. For $m = 2$: 265,338 relations in 136,016,848 attempts. Every count is exact.
- **Metric.** Each curve's yield per attempt ÷ the predicted yield ($(4/3)N^3/n$ for $m = 3$, $2N^2/n$ for $m = 2$).

Result: null supported for both m (frozen rule, PROTOCOL §6; α = 0.05/14 = 0.0036)

test (48 classes)	m = 3: statistic, p	m = 2: statistic, p
T1 Kruskal–Wallis across classes, per-curve yield ÷ predicted	H 44.4, df 47; 0.58	H 43.0, df 47; 0.64
T2 Spearman class yield against $\log_2 H$ (class size)	ρ +0.091; 0.54	ρ +0.144; 0.33
T3 … against $\log_2 f$ (Frobenius conductor)	ρ +0.024; 0.87	ρ +0.029; 0.84
T4 … against number of endomorphism rings	ρ +0.005; 0.97	ρ −0.015; 0.92
T5 … against $\log_2\lvert D_K\rvert$	ρ +0.041; 0.78	ρ +0.117; 0.43
T6 within-class ICC p-values uniform (42 classes with ≥ 2 curves)	KS D 0.090; 0.87	KS D 0.137; 0.38
T7 within-class binomial over-dispersion (pooled Pearson χ^2)	χ^2 331.5, df 284; 0.028	χ^2 296.9, df 284; 0.29

- **Thresholds.** Every test is above α. The smallest p is T7 at $m = 3$ (0.028); it is not significant at the frozen α and is reported as recorded.
- **Positive control.** The planted +20 % arm reads 1.198 ($m = 3$) and 1.200 ($m = 2$), averaged over the 42 classes with at least 2 curves. Both are inside the frozen window [1.12, 1.28]. p2tool's per-class detection rates are 0.89 and 0.92.
- **Spread.** Class means of yield ÷ predicted span 0.938–1.032 for $m = 3$ (mean 0.982, between-class SD 0.019) and 0.944–1.023 for $m = 2$ (mean 0.997, SD 0.018).
- **Strata.** The stratum means are S1 0.986, S2 0.979, S3 0.982 for $m = 3$, and 0.992, 1.002, 0.998 for $m = 2$.
- **Isomorphic models.** The RMS log ratio is 0.048 and 0.044. This is presentation noise: a u-scaling changes which points have small abscissa.

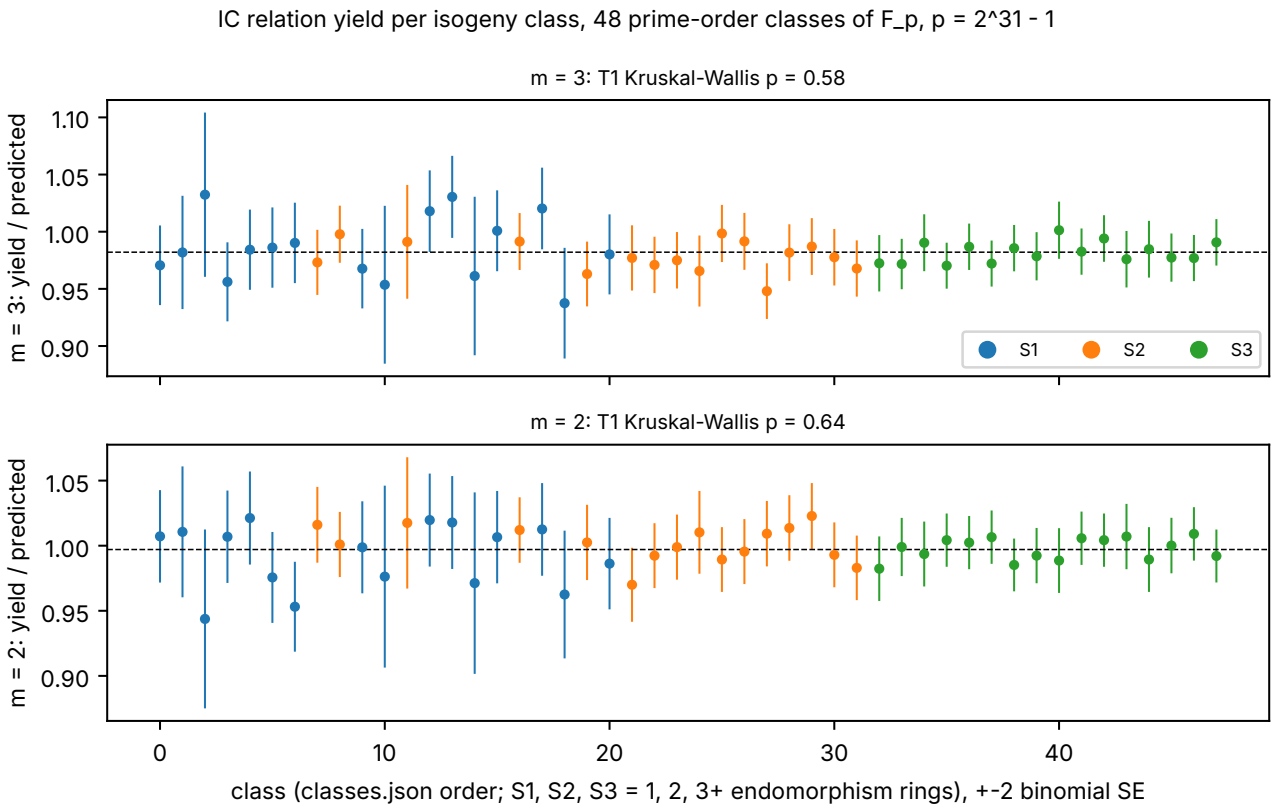


Figure 1: Pooled relation yield ÷ prediction per class, ±2 binomial SE, both m. Recorded values: `phase3/analysis_m{3,2}.json`, `classes`.

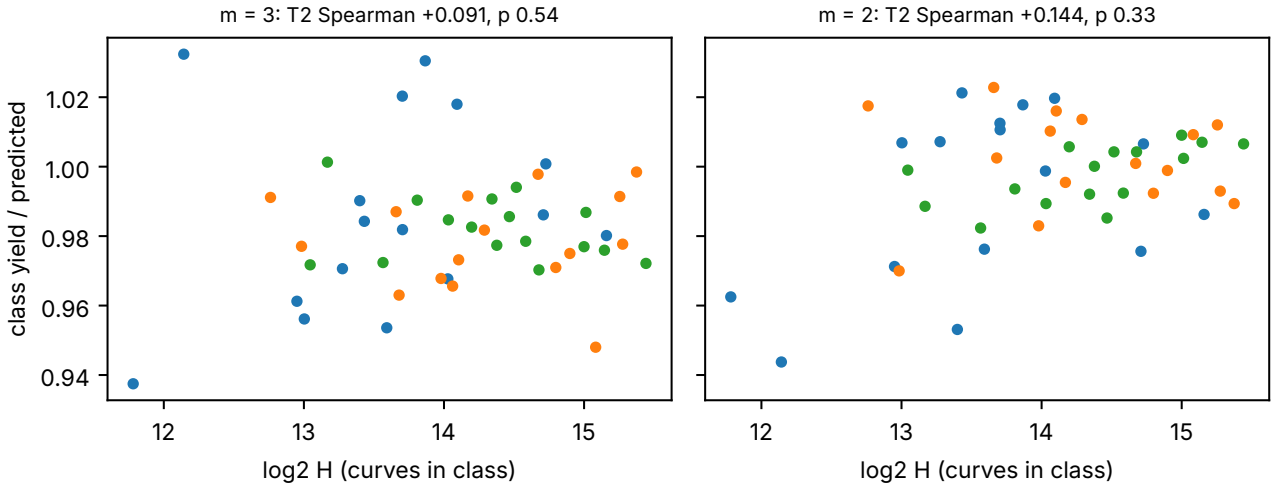


Figure 2: Class yield against class size (T2). There is no association at either m.

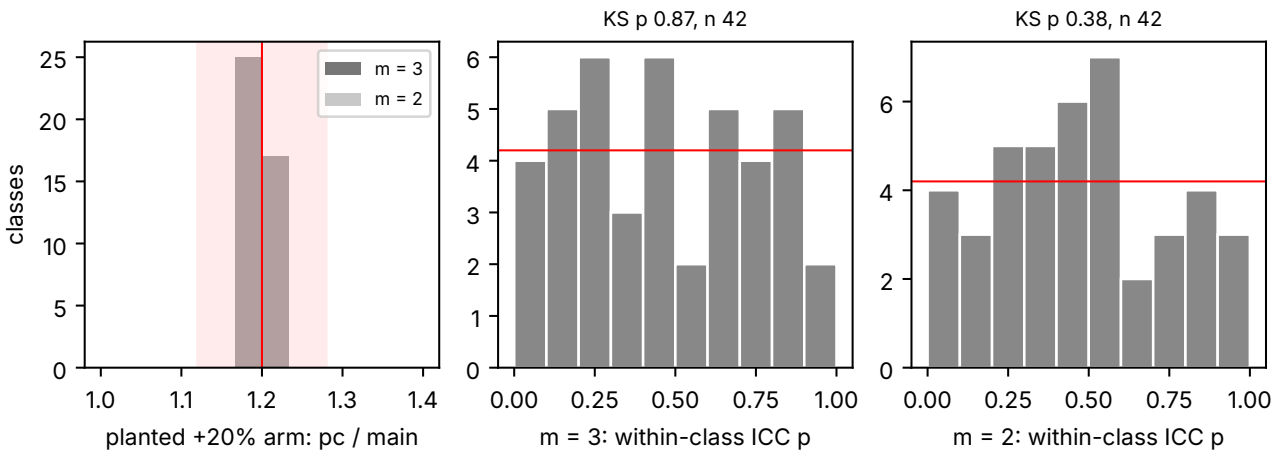


Figure 3: Left: the planted +20 % arm per class, against the expectation 1.2 and the frozen window. Middle and right: per-class within-class ICC permutation p-values, which are uniform under the null.

Reading and limits

- Across 48 prime-order classes of $\mathbb{F}_{2^{31}-1}$, the yield of the prime-abscissa decomposition at $m = 2$ and $m = 3$ is matched by its prediction from $|F|$ and n alone. It does not depend on class size, conductor, number of endomorphism rings or $|D_K|$. Inside a class, the curves along the isogeny walk do not differ. This extends P2′s within-class null (2 classes at 24–32 bits) to 48 classes and every ring structure at 31 bits.
- Combined with phases 1 and 2: over this field, neither rho nor this index-calculus decomposition can tell isogenous curves apart, and neither depends on the class beyond n.
- **Limits.**
  - One factor-base family (smallest abscissa) and one oracle (`mitm`), at $m \leq 3$.
  - About 800 relations per curve, so per-curve resolution is about 3.5 % and class resolution about 1.3–2 %.
  - 42 of the 48 classes have at least 2 curves; S1 walks are horizontal only.
  - Isolation L1/L2 on a shared host. Counts are exact; wall time is unused.
  - The raw sessions are hashed in the benchmark record but not uploaded to S3, because this session has no AWS identity.